
Problem Set 6 - Solutions

Problem 4.5

How much heat is required when 10,000 kg of CaCO_3 is heated at atmospheric pressure from 50 to 880°C ? Use (a) direct integration of the C_p polynomial, (b) ICPH, and (c) MCPH. Report your answers in MJ.

(a) Direct Integration of the C_p Polynomial

```
In[1]:= a = 12.572; (*Table C.2 page 670*)  
b = 2.637 * 10^-3;  
c = 0;  
d = -3.12 * 10^5;
```

```
In[5]:= Cp = a + b * T + d * T^-2; (*Equation 4.5*)
```

```
In[6]:= T0 = 50 + 273.15; (*K*)  
T1 = 880 + 273.15;  
R = 8.314; (* J / (mol * K) *)
```

```
In[9]:= Q = R * Integrate[Cp dT, {T, T0, T1}] (*Units come from R - J/mol*)
```

```
Out[9]= 94408.99
```

```
In[10]:= Q * J / mol * 1 mol / 100.087 g * 10000 kg * 1000 g / kg * 1 MJ / 10^6 J (*ANS*)
```

```
Out[10]= 9432.693 MJ
```

9432.69 MJ of heat energy is required. //ANS

(b) ICPH, Eq. 4.8

```
In[11]:= ICPH = a * (T1 - T0) + b / 2 * (T1^2 - T0^2) + c / 3 * (T1^3 - T0^3) + d * (T1 - T0) / (T1 * T0)
```

```
Out[11]= 11355.42
```

```
In[12]:= Q = R * ICPH
```

```
Out[12]= 94408.99
```

$$\text{In}[13]:= Q * \frac{\text{J}}{\text{mol}} * \frac{1 \text{ mol}}{100.087 \text{ g}} * 10000 \text{ kg} * \frac{1000 \text{ g}}{\text{kg}} * \frac{1 \text{ MJ}}{10^6 \text{ J}} (*\text{ANS}*)$$

Out[13]=

9432.693 MJ

9432.69 MJ, same as part (a). //ANS

(c) MCPH, Eq. 4.9

$$\text{In}[14]:= \text{MCPH} = a + \frac{b}{2} * (T1 + T0) + \frac{c}{3} (T1^2 + T0^2 + T1 * T0) + \frac{d}{T1 * T0}$$

Out[14]=

13.68123

$$\text{In}[15]:= Q = R * \text{MCPH} * (T1 - T0)$$

Out[15]=

94408.99

$$\text{In}[16]:= Q * \frac{\text{J}}{\text{mol}} * \frac{1 \text{ mol}}{100.087 \text{ g}} * 10000 \text{ kg} * \frac{1000 \text{ g}}{\text{kg}} * \frac{1 \text{ MJ}}{10^6 \text{ J}} (*\text{ANS}*)$$

Out[16]=

9432.693 MJ

9432.69 MJ, same as part (a). //ANS

Problem 4.9

A process stream is heated as a gas from 25 to 250°C at constant P. A quick estimate of the energy requirement is obtained from Eq. 4.3, with C_p taken as constant and equal to its value at 25°C. Is the estimate of Q likely to be low or high? Why?

Solution

This question is answered with a calculation. While the fluid identity is unspecified, we can use a representative ideal gas, such as nitrogen. Nitrogen appears in the plot in Figure 4.1 in the textbook showing heat capacity as a function of temperature, but not the temperature range given in this problem, so we will re-build this plot. Use Table C.1 on page 656 along with Eq. 4.3 to construct, plot, and integrate the heat capacity polynomial.

It is important to point out that the sample calculation shown here does not represent a general solution to this answer, but is only illustrative.

$$\text{In}[17]:= a = 3.280; (*\text{Table C.1}*)$$

$$b = 0.593 / 1000;$$

$$d = 0.040 / 10^{-5};$$

$$\text{In}[20]:= C_p = a + b * T + d * T^{-2}; (*\text{Equation 4.5}*)$$

```
In[21]:= T1 = 25 + 273.15; (*K*)
          T2 = 250 + 273.15;
          R = 8.314; (*  $\frac{\text{J}}{\text{mol}\cdot\text{K}}$  *)
```

```
In[24]:= Q = R *  $\int_{T1}^{T2} C_p dT$  (*  $\frac{\text{J}}{\text{mol}}$  *)
```

```
Out[24]= 6639.237
```

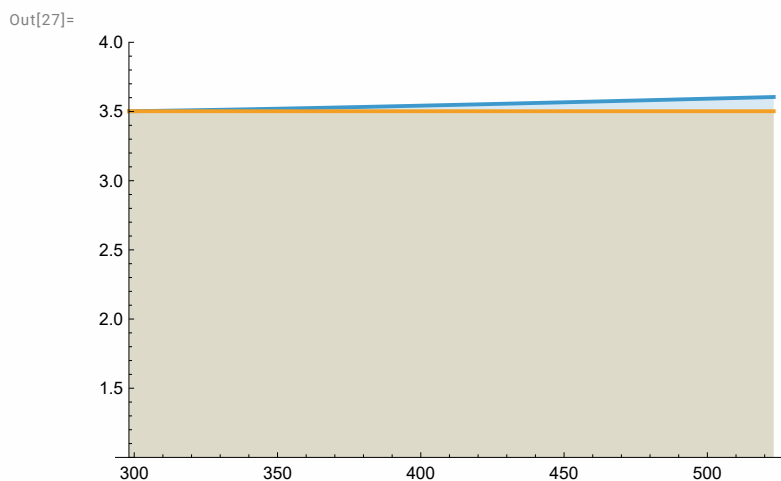
```
In[25]:= Cp298 = 3.502; (*Table 3.1 in  $\frac{\text{J}}{\text{mol K}}$  *)
```

```
In[26]:= R * Cp298 * (250 - 25) (*  $\frac{\text{J}}{\text{mol}}$  *)
```

```
Out[26]= 6551.016
```

As can be seen in these two results, the heat estimate from the constant C_p is low. The estimate is low because the heat capacity increases with temperature. This can be seen in Figure 4.1 in the textbook. Using a constant heat capacity taken from the left side of the plot will underestimate the value of the area under the curve. Since the heat is equal to the integral of the curve, underestimating the area under the curve will underestimate the heat. This is illustrated in the plot below. The bluish-amber shaded area represents the integral of the heat capacity assuming a constant C_p . The blue triangle represents the difference in the integrated areas.

```
In[27]:= Plot[{Cp, Cp298}, {T, T1, T2}, PlotRange -> {1, 4}, Filling -> Bottom]
```



The reason that the heat capacity increases with temperature is that as temperature is increased, additional vibrational and rotational modes are available to the molecules to store heat energy. Note that this does not happen with molecules like argon (see Fig. 4.1). This is because the argon molecule is monatomic and spherical in shape and does not vibrate or rotate the same way that the polyatomic molecules do. You can see this if you think about the chemical bond as a spring with two masses attached to the ends described with classical mechanics. The spring can stretch and compress in a manner similar to the chemical bond. These stretches and compressions act as a reservoir for thermal energy.

Problem 4.10

- (a) Evaluate the latent heat of vaporization ΔH_n of n-Pentane by Eq. 4.13. How does this result compare with the value listed in Table B.2?
- (b) Handbook values for latent heats of vaporization at 25°C of four compounds are given in the table below. Calculate ΔH_n by Eq. 4.14, and compare the results with the values given in Table B.2.

Latent heats of vaporization at 25 °C in J/g

n-Pentane	366.3	Benzene	433.3
n-Hexane	366.1	Cyclohexane	392.5

Solution - Part (a) - Latent Heats by Eq. 4.13 (Reidel's Equation)

```
In[28]:= (*n-Pentane*)
Tn = 309.2; (*K, Table B.1 page 663*)
Tc = 469.7; (*K, Table B.1 page 663*)
Trn = Tn / Tc;
Pc = 33.70; (*bar*)
R = 8.314; (* J / (mol * K) *)
ΔHn = R * Tn * (1.092 * (Log[Pc] - 1.013) / (0.930 - Trn)) (*J/mol*)
```

```
Out[33]=
25 875.65
```

The latent heat of vaporization ΔH_n of n-Pentane by Eq. 4.13 is 25,875.65 J/mol. //ANS

The value given in Table B.2 is 25.79 kJ/mol.

```
In[*]:= .001 ΔHn - 25.79
          25.79 * 100 (*percent error*)
```

```
Out[*]=
0.3320947
```

Comparison of the calculated ΔH_n with the value given in Table B.2 shows 0.33% error. //ANS

Solution - Part (b) - Latent Heats by Eq. 4.14 (Watson's Equation)

This solution uses "lists" to answer the question for all four compounds simultaneously. In the work that follows, the list order is {n-Pentane, n-Hexane, Benzene, Cyclohexane}.

```
ΔH1 = {366.3, 366.1, 433.3, 392.5}; (*Given in J/g at 25°C*)
T1 = 298.15; (*25°C Given, convert to K*)
T2 = {309.2, 341.9, 353.2, 353.9}; (*Tn in K from App B pp. 663-665*)
Tc = {469.7, 507.6, 562.2, 553.6}; (*T2 in K from App B pp. 663-665*)
Tr2 = T2 / Tc;
Tr1 = T1 / Tc;
```

```

In[*]:= ΔH2 = ΔH1 *  $\left(\frac{1 - Tr2}{1 - Tr1}\right)^{0.38}$  (*in J/g*)
Out[*]=
{357.1486, 334.9128, 396.4638, 357.4432}

In[*]:= MW = {72.150, 86.177, 78.114, 84.160}; (*App B pp. 663-665*)
ΔH2pm = ΔH2 * MW / 1000 (*kJ/mol where "pm" is "per mol"*)
Out[*]=
{25.76827, 28.86178, 30.96937, 30.08242}

```

The latent heats are shown above in kJ/mol. //ANS

The Table B.2 values in kJ/mol are listed in ΔHn:

```

In[*]:= ΔHn = {25.79, 28.85, 30.72, 29.97}; (*Table B.2 in kJ/mol.*)

In[*]:=  $\frac{\Delta Hn - \Delta H2pm}{\Delta Hn} * 100$  (*percent error*)
Out[*]=
{0.08425029, -0.04083255, -0.8117604, -0.3751171}

```

The percent errors are shown above. //ANS

Problem 4.12

Handbook values for the latent heat of vaporization in J/g are given in the table for several pure liquids at 0°C.

	ΔH at 0°C
Chloroform	270.9
Methanol	1,189.5
Tetrachloromethane	217.8

Calculate:

(a) The value of the latent heat at T_n by Eq. 4.14, given the value listed at 0 °C.

(b) The value of the latent heat at T_n by Eq. 4.13.

By what percentage do these results differ from the value listed in Table B.2 of App. B?

Solution - Part (a) - Latent Heats by Eq. 4.14 (Watson's Equation)

The solutions in Parts (a) and (b) use "lists" to generate answers for all three compounds simultaneously. The list order is {chloroform, methanol, tetrachloromethane}.

```

In[*]:= ΔH1 = {270.9, 1189.5, 217.8}; (*in J/g, given*)
T1 = 273.15; (*K, given as 0°C*)
T2 = {334.3, 337.9, 349.8}; (*Tn in K from Table B.1*)
Tc = {536.4, 512.6, 556.4}; (*in K from Table B.1*)
Tr2 = T2 / Tc;
Tr1 = T1 / Tc;

```

```

In[*]:= ΔH2 = ΔH1 *  $\left(\frac{1 - Tr2}{1 - Tr1}\right)^{0.38}$  (*J/g*)
Out[*]=
{ 245.0102, 1 055.198, 193.1892 }

In[*]:= MW = {119.377, 32.042, 153.822}; (*from Table B.1*)
ΔH2pm = ΔH2 * MW / 1000 (*kJ/mol*)
Out[*]=
{ 29.24859, 33.81067, 29.71675 }

```

The latent heats are shown above in kJ/mol. //ANS

The Table B.2 values in kJ/mol are listed in ΔHn:

```

In[*]:= ΔHn = {29.24, 35.21, 29.82}; (*ΔHn from Table B.2 in kJ/mol*)

In[*]:=  $\frac{\Delta Hn - \Delta H2pm}{\Delta Hn} * 100$  (*percent error*)
Out[*]=
{ -0.029 365 08, 3.974 251, 0.346 251 }

```

The percent errors are shown above. //ANS

Solution - Part (b) - Latent Heats by Eq. 4.13 (Reidel's Equation)

```

In[*]:= (*Pc is in App B pp. 663-665*)
Pc = {54.72, 80.97, 45.60}; (*bar*)
R = 8.314; (* $\frac{J}{mol \cdot K}$  from App A.*)
ΔH2 = R * T2 *  $\frac{1.092 * (\text{Log}[Pc] - 1.013)}{0.930 - Tr2} * \frac{1}{1000}$  (*kJ/mol*)
Out[*]=
{ 29.57426, 38.3009, 29.58414 }

```

The latent heats are shown above in kJ/mol. //ANS

```

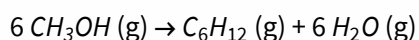
In[*]:=  $\frac{\Delta Hn - \Delta H2}{\Delta Hn} * 100$  (*percent error*)
Out[*]=
{ -1.143 155, -8.778 465, 0.790 938 4 }

```

The percent errors are shown above. //ANS

Problem 4.20

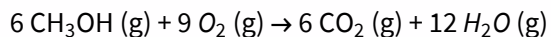
Hydrocarbon fuels can be produced from methanol by reactions such as the following, which yields 1-hexene:



Compare the standard heat of combustion at 25°C of 6 CH_3OH (g) with the standard heat of combustion at 25°C of C_6H_{12} (g) for reaction products CO_2 (g) and H_2O (g).

Solution

Six molecules of Methanol



(*Table C.4 values: CO_2 : -393509 $\text{H}_2\text{O (g)}$: -241818 $\text{CH}_3\text{OH (g)}$: -200660*)

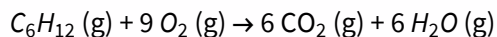
$\Delta H_m = (-393\,509) * 6 + (-241\,818) * 12 - (-200\,660) * 6$ (*//ANS, J per 6 moles of meoh*)

Out[*]=

-4 058 910

The standard heat of combustion at 25°C is -4,058,910 Joules per six moles of $\text{CH}_3\text{OH (g)}$. //ANS

One molecule of 1-Hexene



(*Table C.4 values: $\text{C}_6\text{H}_{12} \text{ (g)}$: -41950*)

In[*]:= $\Delta H_h = (-393\,509) * 6 + (-241\,818) * 6 - (-41\,950) * 1$ (*//ANS J/mol*)

Out[*]=

-3 770 012

The standard heat of combustion at 25°C is -3,770,012 Joules per mole of $\text{C}_6\text{H}_{12} \text{ (g)}$. //ANS

In[*]:= $\Delta H_m - \Delta H_h$

Out[*]=

-288 898

The comparison is by difference, and we see that the combustion of 6 CH_3OH releases more heat than one C_6H_{12} at standard conditions. The heat of combustion of CH_3OH is lower than that of C_6H_{12} by -288,898 J/mol. //ANS